MODERNIZATION EQUIPMENT ACQUISITION PROGRAM

Test and Evaluation Master Plan (TEMP)

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Purpose and Scope

This Test and Evaluation Master Plan (TEMP) defines the integrated test and evaluation strategy for the Modernization Equipment Acquisition (MEA) program. It establishes the sequence, entry criteria, success criteria, and responsibilities for all formal test events from subsystem acceptance testing through system integration test (SIT) and operational assessment. This document governs test planning and execution for all four program subsystems: Propulsion, Navigation, Communications, and Power Distribution.

This TEMP does not govern developmental testing conducted internally by contractors prior to government-witnessed events. Contractor developmental test plans are contractually required deliverables and shall be submitted to the COR no later than 90 days after contract award for each subsystem.

Test Event Sequence

Formal test events shall occur in the following sequence. Subsystem Acceptance Tests (SATs) for each subsystem shall be completed before System Integration Test entry. No subsystem may proceed to SIT without a signed SAT completion memo from the cognizant COR and a written determination from the Program Manager that all SAT entry and exit criteria have been satisfied.

Phase 1 — Subsystem Acceptance Tests (SAT) Each subsystem undergoes a standalone acceptance test conducted at the contractor's facility or a government-designated test facility. SATs verify that individual subsystems meet their contractual technical requirements prior to integration.

Phase 2 — System Integration Test (SIT) SIT verifies that all four subsystems operate correctly as an integrated system. SIT is conducted at the government-designated integration facility. SIT entry requires successful completion of all four SATs and satisfaction of all SIT entry criteria defined in Section 4.

Phase 3 — Operational Assessment (OA) Following successful SIT completion, the integrated system undergoes an operational assessment conducted by end users under realistic operational conditions. OA findings inform the final fielding decision.

Subsystem Acceptance Test Requirements

3.1 Propulsion Subsystem SAT

The Propulsion SAT shall demonstrate sustained operation of the turbine assembly across the full operational envelope defined in the Propulsion Subsystem Performance Specification (MEA-SPEC-PROP-001). Required test demonstrations include: cold-start performance at minimum rated temperature, sustained operation at maximum rated output for no less than four continuous hours, emergency shutdown and restart cycle, and turbine seal integrity verification under maximum pressure conditions. All turbine seal integrity test runs shall return zero leakage measurements. Any leakage detected during seal integrity testing constitutes an automatic test failure requiring root cause analysis and retest before SAT can be signed off.

The Propulsion SAT shall be government-witnessed. Apex Defense Systems shall provide a minimum 30-day advance notice of SAT readiness. The government test team shall have the right to observe all test runs and to call additional test points within the approved test matrix if anomalies are observed.

3.2 Navigation Subsystem SAT

The Navigation SAT shall verify position accuracy, acquisition time, and GPS/INS fusion performance in accordance with the Navigation Subsystem Performance Specification (MEA-SPEC-NAV-001). Required test demonstrations include: GPS-denied navigation accuracy over a minimum 60-minute period, position fix acquisition time from cold start, INS drift rate measurement over a four-hour period, and firmware update installation and verification. All firmware loaded on the navigation unit at the time of SAT shall be the approved production release version. Pre-production or developmental firmware versions are not acceptable for SAT execution.

3.3 Communications Subsystem SAT

The Communications SAT shall verify waveform performance, link establishment time, and interoperability with designated external systems in accordance with the Communications Subsystem Performance Specification (MEA-SPEC-COMM-001). Required test demonstrations include: link establishment within the maximum specified time across all supported waveforms, data throughput at minimum specified rates, interoperability verification with at least two designated external systems, and graceful degradation testing under simulated jamming conditions.

3.4 Power Distribution Subsystem SAT

The Power Distribution SAT shall verify load distribution performance, fault isolation, and capacitor bank charge/discharge characteristics in accordance with the Power Distribution Subsystem Performance Specification (MEA-SPEC-PWR-001). Required test demonstrations include: full-load distribution across all circuit branches simultaneously, fault isolation response time within the maximum specified threshold, capacitor bank charge time from minimum state of charge, and thermal performance under sustained maximum load.

System Integration Test Entry Criteria

All of the following entry criteria must be satisfied before the program may enter System Integration Test. The Program Manager shall certify in writing that all entry criteria are met before SIT activities commence. Partial satisfaction of entry criteria does not constitute SIT readiness.

4.1 All four Subsystem Acceptance Tests shall be complete with no open SAT failure reports. SAT failure reports that have been formally closed with government concurrence are acceptable; open or conditionally closed failure reports are not.

4.2 All subsystem hardware shall be delivered to the integration facility and confirmed receipt by the integration facility manager. No SIT activity may begin on a subsystem whose hardware has not been physically received and inventoried at the integration facility.

4.3 The System Integration Test Procedure (MEA-SITP-001) shall be approved by the Program Manager and the Government Test Agency no later than 30 days before SIT entry. Any unapproved or draft procedures may not be used to conduct government-witnessed test events.

4.4 All software and firmware loaded on subsystems at SIT entry shall be configuration-controlled release versions. Version identification for all software and firmware shall be documented in the SIT Configuration Baseline Record, submitted to the COR no later than 14 days before SIT entry.

4.5 The integration facility shall have completed facility readiness certification within 60 days of planned SIT entry. An expired or incomplete facility readiness certification shall constitute a SIT entry blocker.

4.6 All open safety-critical findings from subsystem acceptance tests shall be formally closed with documented root cause and corrective action. Non-safety-critical findings may remain open at SIT entry provided they are documented in the SIT open items log and dispositioned as "monitor" or "fix post-SIT" with Program Manager approval.

System Integration Test Success Criteria

SIT shall be considered successful when all of the following conditions are met. Partial success is not defined for SIT — the test event either satisfies all success criteria or is considered a failure requiring disposition by the Program Manager and Government Test Agency before an OA date can be confirmed.

5.1 All test procedures in the approved System Integration Test Procedure shall be executed with a pass result, or with a documented deviation approved in writing by the Program Manager and Government Test Agency prior to execution.

5.2 All four subsystems shall demonstrate interoperability across defined interfaces with no unresolved interface failures at the conclusion of the SIT period.

5.3 No safety-critical anomalies shall remain open at SIT completion.

5.4 The integrated system shall complete a minimum 72-hour continuous operation demonstration with no interruptions attributable to subsystem failures.

Roles and Responsibilities

Program Manager: Accountable for overall test program execution. Certifies SIT entry criteria are met. Approves deviations from approved test procedures. Makes final disposition on SIT failure reports.

Contracting Officer's Representative (COR): Witnesses government-witnessed test events. Signs SAT completion memos. Receives and reviews contractor test reports. Tracks open test failure reports.

Government Test Agency: Develops and approves the System Integration Test Procedure in coordination with the Program Office. Provides test conductors for government-witnessed events. Prepares the official SIT report.

Contractors: Conduct developmental testing prior to government-witnessed events. Provide 30-day advance notice of SAT readiness. Submit test reports within 14 days of each test event completion. Support Government Test Agency during SIT.

Schedule

The following test schedule reflects the program baseline as of the date of this document. Schedule updates resulting from program events shall be reflected in quarterly program status reports and in updated versions of this TEMP.

Test Event	**Subsystem**	**Planned Start**	**Planned Completion**
Propulsion SAT	Propulsion	November 2024	December 2024

Test Event	**Subsystem**	**Planned Start**	**Planned Completion**
Navigation SAT	Navigation	November 2024	December 2024
Communications SAT	Communications	December 2024	January 2025
Power Distribution SAT	Power Distribution	December 2024	January 2025
System Integration Test	All	March 2025	May 2025
Operational Assessment	All	July 2025	September 2025

Known Risks to Test Schedule

At the time of this TEMP version, two program risks are assessed as having potential impact to the test schedule.

The Propulsion subsystem turbine seal deficiency, currently tracked as Risk R-001 at Red status, directly affects the Propulsion SAT. Specifically, the turbine seal integrity verification required under Section 3.1 of this TEMP cannot be passed while the turbine seal deficiency remains unresolved. If the deficiency is not corrected prior to the planned November 2024 SAT start, the Propulsion SAT will slip, which will in turn delay SIT entry under entry criterion 4.1. The current program baseline reflects a March 2025 SIT entry date, which already incorporates a six-week schedule reserve to accommodate resolution of this risk. Additional slip beyond the current corrective action plan timeline would require a formal schedule revision.

The Navigation subsystem firmware timing conflict, tracked as Risk R-002 at Yellow status, requires delivery and verification of a production firmware patch prior to Navigation SAT execution under Section 3.2 of this TEMP, which requires that only approved production firmware be loaded at SAT. Meridian Technologies has committed to patch delivery in Q1 FY2025. Provided this commitment is met, no impact to the SIT schedule is anticipated.

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Prepared by: MEA Test and Evaluation Division **Reviewed by:** Government Test Agency Representative **Approved by:** Program Manager